



Overview of Diabetes Mellitus

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Diabetes mellitus is a metabolic disease characterized by elevated blood glucose levels and inadequate insulin release relative to the body's needs.

Between 11 and 14% of adults worldwide have diabetes ([1](#), [2](#)). There are 2 main types of diabetes mellitus (diabetes), [type 1 diabetes](#), which comprises 5 to 10% of all diagnosed diabetes, and [type 2 diabetes](#), which comprises 90 to 95% of diabetes cases ([3](#)). Other, less prevalent types account for the remainder of diabetes cases.

Overview of Diabetes

VIDEO

DIABETIC KETOACIDOSIS (DKA)

Treatment

- FLUIDS for DEHYDRATION
- INSULIN to lower

General references

1. [International Diabetes Federation](#). About diabetes: Facts and Figures. Accessed November 30, 2025.
2. [World Health Organization](#). Diabetes: Key Facts. November 14, 2024. Accessed November 30, 2025.
3. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S27-S49. doi:10.2337/dc25-S002

Types of Diabetes

Type 1 and 2 Diabetes

Type 1 diabetes is primarily characterized by inadequate insulin production due to autoimmune destruction of pancreatic beta cells, whereas type 2 diabetes is primarily characterized by insulin resistance and a relative insufficiency of insulin. The 2 types of diabetes can be distinguished by a number of features (see table [General Characteristics of Types 1 and 2 Diabetes Mellitus](#)). There is, however, significant overlap between type 1 and type 2 diabetes. Terms that describe the age of onset (juvenile or adult) or type of treatment (insulin-dependent or non-insulin-dependent) are not used because of overlap in age groups and treatments between disease types.

Prediabetes, also called impaired glucose regulation or impaired glucose tolerance, is an intermediate, possibly transitional, state between normal glucose metabolism and diabetes mellitus, most commonly [type 2 diabetes](#) (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)).

TABLE		
General Characteristics of Types 1 and 2 Diabetes Mellitus		
Characteristic	Type 1	Type 2
Age at onset	Most commonly < 30 years	Most commonly > 30 years
Associated obesity	Uncommon	Very common
Propensity to ketoacidosis requiring insulin treatment for control	Yes	No
Plasma levels of endogenous insulin	Extremely low to undetectable	Variable; may be low, normal, or elevated depending on degree of insulin resistance and insulin secretory defect
Twin concordance	≤ 50%	> 90%
Associated with specific HLA-D antigens	Yes	No
Pancreatic autoantibodies at diagnosis	Yes, but may be absent	No
Islet pathology	Insulitis, selective loss of most beta cells	Smaller, normal-appearing islets; amyloid (amylin) deposition common

Characteristic	Type 1	Type 2
Prone to develop diabetic complications (retinopathy, nephropathy, neuropathy, atherosclerotic cardiovascular disease)	Yes	Yes
Hyperglycemia responds to non-insulin antihyperglycemic medications	No	Yes, initially in many patients
HLA = Human leukocyte antigen.		

Other Types and Causes of Diabetes

Other types of diabetes mellitus account for a smaller proportion of cases. Causes include:

- Gestational diabetes
- Monogenic diabetes
- Latent autoimmune diabetes in adults
- [Cystic fibrosis-related diabetes](#)
- Pancreatogenous diabetes (due to destruction or removal of pancreas, including chronic [pancreatitis](#), sometimes referred to as type 3c diabetes)
- Other conditions that affect the pancreas (eg, [hemochromatosis](#))
- Post-transplant diabetes
- Malnutrition-related diabetes
- Endocrinopathies (eg, [Cushing syndrome](#), [acromegaly](#))
- Medications, most notably glucocorticoids, beta-blockers, protease inhibitors, atypical antipsychotics, immune checkpoint inhibitors, and calcineurin inhibitors

Gestational diabetes

[Gestational diabetes](#) occurs in some pregnant people because pregnancy induces insulin resistance.

Monogenic diabetes

Monogenic forms of diabetes are caused by genetic defects affecting beta-cell function, insulin action, or mitochondrial DNA (eg, neonatal diabetes).

Latent autoimmune diabetes in adults

Latent autoimmune diabetes is a variant that develops in adulthood in which one or more autoantibodies is present. It is more slowly progressive than classic type 1 diabetes, and some adults do not need insulin when dysglycemia first develops. This form of diabetes is commonly misdiagnosed as type 2 diabetes due to its slower progression and overlapping clinical features.

Diagnosis of Diabetes Mellitus

Diabetes mellitus is suggested by typical symptoms and signs and confirmed by measurement of plasma glucose ([1](#), [2](#)). It is often detected through screening.

Diabetes is diagnosed when one of the following is present:

- Glycosylated hemoglobin (HbA1C) $\geq 6.5\%$ (≥ 48 mmol/mol)
- Fasting plasma glucose (FPG) level ≥ 126 mg/dL (≥ 7.0 mmol/L)
- 2-hour glucose tolerance test (OGTT) ≥ 200 mg/dL (≥ 11.1 mmol/L)
- Random glucose ≥ 200 mg/dL (≥ 11.1 mmol/L) with symptoms of [hyperglycemic or hypoglycemic crisis](#)

The diagnosis of diabetes is classified into [type 1 diabetes](#), [type 2 diabetes](#), [gestational diabetes](#), or other forms of diabetes based on the patient presentation as well as genetic, immunologic, and contextual factors ([1](#)).

HbA1C is a form of hemoglobin that is chemically attached to a sugar that increases with blood glucose and has a validated relationship with average glucose level over the preceding 3 months. HbA1C measurements are included in the diagnostic criteria for diabetes:

- HbA1C $\geq 6.5\%$ (≥ 48 mmol/mol) = diabetes
- HbA1C 5.7 to 6.4% = prediabetes or at risk of diabetes

However, HbA1C is an indirect measure of blood glucose; values may be falsely high or low (see [Monitoring](#)) and can vary with race or ethnicity. Tests must be performed in a certified clinical laboratory with an assay that is certified and standardized to a reference assay. Point-of-care fingerstick HbA1C measurements should not be used for diagnostic purposes, although they can be used for monitoring diabetes control.

Measurement after an 8- to 12-hour fast (**FPG**) or 2 hours after ingestion of a concentrated glucose solution (OGTT) is preferred (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)).

OGTT involves measuring plasma glucose 2 hours after a 75-g glucose load dissolved in liquid. It is more sensitive for diagnosing diabetes and impaired glucose tolerance but is less convenient and reproducible than FPG. It is therefore rarely used routinely, except for diagnosing [gestational diabetes](#) and for research purposes.

In practice, diabetes mellitus or impaired fasting glucose regulation is often diagnosed using random measures of plasma glucose or of HbA1C. A random glucose value > 200 mg/dL (> 11.1 mmol/L) may be diagnostic, but values can be affected by recent meals and must be confirmed by repeat testing; testing twice may not be necessary in the presence of symptoms of diabetes.

In sick, hospitalized patients, blood glucose may be elevated due to stress hyperglycemia. Patients with elevated plasma glucose levels during a hospitalization should be evaluated for diabetes once they are stable and outside of the hospital.

Urine glucose measurement, once commonly used, is no longer used for diagnosis or monitoring because it is neither sensitive nor specific.

Pearls & Pitfalls

- Point-of-care fingerstick HbA1C tests are not accurate enough to be used for initial diagnosis of diabetes.

Prediabetes is diagnosed when the HbA1C, OGTT, or FPG is between the normal range and the range diagnostic for diabetes. (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)).

TABLE

Diagnostic Criteria for Diabetes Mellitus and Prediabetes*			
Test	Normal	Prediabetes (Impaired Glucose Regulation)	Diabetes
FPG (mg/dL [mmol/L])	< 100 (< 5.6)	100–125 (5.6–6.9)	≥ 126 (≥ 7.0)
OGTT (mg/dL [mmol/L])	< 140 (< 7.8)	140–199 (7.8–11.0)	≥ 200 (≥ 11.1)
HbA1C (%)	< 5.7	5.7–6.4	≥ 6.5
Random glucose (mg/dL [mmol/L])	< 200 (< 11.1)	—	> 200 (> 11.1) in a patient with symptoms

* See also [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S27-S49. doi:10.2337/dc25-S002

FPG = fasting plasma glucose; HbA1C = glycosylated hemoglobin; OGTT = oral glucose tolerance test, 2-hour glucose level.

CLINICAL CALCULATORS

[Glycemic Assessment](#)



Diagnosis reference

1. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(Supplement_1):S27-S49. doi:10.2337/dc25-S002

Complications of Diabetes Mellitus

Acute complications of diabetes include metabolic derangements such as [diabetic ketoacidosis](#), [hypoglycemia](#), and [hyperosmolar hyperglycemic state](#).

[Long-term complications](#), due to chronic hyperglycemia, include primarily vascular complications that affect small vessels (microvascular), large vessels (macrovascular), or both.

Microvascular disease underlies 3 common and severe complications of diabetes mellitus:

- [Retinopathy](#)
- [Nephropathy](#)
- [Neuropathy](#)

Microvascular disease may also impair wound healing, so that even minor breaks in skin integrity can develop into deeper ulcers and easily become infected, particularly in the lower extremities. Intensive control of plasma glucose can prevent or delay many of these complications but will not reverse them once established.

Macrovascular disease involves [atherosclerosis](#) of large vessels, which can lead to [angina pectoris](#) and [myocardial infarction](#), [transient ischemic attacks](#) and [strokes](#), and [peripheral artery disease](#).

Immune dysfunction is another major complication and develops from the direct effects of hyperglycemia on [cellular immunity](#). Patients with diabetes mellitus are particularly susceptible to bacterial and fungal infections.

Special Populations

Diabetes care requires careful adjustment for patient factors, including those related to age and lifestyle, comorbid conditions, and need for treatment of other acute or chronic conditions.

Patients with difficulty maintaining goal glucose levels

The term brittle diabetes has been used to refer to patients who have dramatic, recurrent swings in glucose levels, often for no apparent reason. Labile plasma glucose levels are more likely to occur in patients with type 1 diabetes because endogenous insulin production is almost completely absent, and in some patients, counter-regulatory response to hypoglycemia is impaired. Other causes of labile plasma glucose levels include occult infection, gastroparesis (which leads to erratic absorption of dietary carbohydrates), and other endocrine disorders (eg, [Addison disease](#)).

Patients with chronic difficulty maintaining acceptable glucose levels should be evaluated for situational factors that affect glucose control. Such factors include inadequate patient education or understanding that leads to errors in insulin administration, suboptimal food choices, and psychosocial stress that expresses itself in inconsistent medication use and food intake.

The initial approach is to thoroughly review self-care techniques, including insulin preparation and injection and glucose testing. Increased frequency of self-testing may reveal previously unrecognized patterns and provides the patient with helpful feedback. A thorough dietary history, including timing of meals, should be taken to identify potential contributions to poor control. Underlying disorders, such as [Cushing syndrome](#), should be excluded by physical examination and appropriate laboratory tests.

For some patients being treated with insulin, adjusting to a more intensive regimen that allows for frequent dose adjustments (based on glucose testing) is helpful. Continuous glucose monitoring with alarms and sensor-augmented or hybrid closed-loop insulin pump therapy are useful tools in individuals who fluctuate between hypoglycemia and hyperglycemia.

Children and adolescents

[Diabetes in children and adolescents](#) is discussed in detail elsewhere.

Patients with heart failure

Patients with type 2 diabetes who are hospitalized for [heart failure](#) should receive sodium glucose cotransporter 2 inhibitors (SGLT-2 inhibitors) after the acute illness has improved ([1](#)). However, SGLT2 inhibitors should be avoided in patients with severe illness, those who are fasting, or those with ketonemia or ketonuria because these conditions increase the risk of SGLT2 inhibitor-induced [diabetic ketoacidosis](#). Thiazolidinediones, such as pioglitazone, should also be avoided in patients with heart failure.

Hospitalized patients

Hospitalization for diabetes

Diabetes mellitus may be a primary reason for hospitalization or may accompany other illnesses that require inpatient care. Patients with [hypoglycemia](#) induced by sulfonylureas, poorly controlled hyperglycemia, or acute worsening of diabetic complications may benefit from brief hospitalization.

Hospitalized patients with pre-existing diabetes

When other illnesses mandate hospitalization, some patients can continue on the diabetes treatment regimens they use at home. However, glucose control often proves difficult, and it is often neglected when other diseases are more acute. Restricted physical activity and acute illness worsen hyperglycemia in some patients, whereas dietary restrictions and symptoms that accompany illness (eg, nausea, vomiting, diarrhea, anorexia) precipitate hypoglycemia in others—especially when antihyperglycemic medication doses remain unchanged. In addition, it may be difficult to control glucose adequately in patients who are hospitalized because usual routines (eg, timing of meals, medications, and procedures) are inflexibly timed relative to diabetes treatment regimens.

In patients who are hospitalized, oral antihyperglycemic medications often need to be stopped. Metformin can cause [lactic acidosis](#) in patients with end-stage renal disease and has to be stopped if contrast agents need to be given. Therefore, metformin is withheld in all but the most stable patients who are hospitalized. Sulfonylureas can cause hypoglycemia and should also be stopped.

Most inpatients can be appropriately treated with basal insulin without or with supplemental short-acting insulin.

Sliding-scale insulin should not be the only intervention to correct hyperglycemia; it is reactive rather than proactive, and it leads to poor glycemic control compared to basal-bolus insulin. Longer-acting insulins should be adjusted to prevent hyperglycemia rather than just using short-acting insulins to correct it.

Patients using insulin pumps should continue those in the hospital when feasible and appropriate, with confirmatory blood glucose measurements. Patients who are already monitoring glucose with continuous glucose monitoring, especially those at risk for hypoglycemia, should continue to use their devices in conjunction with point-of-care fingerstick glucose monitoring for confirmation of hypoglycemia and for insulin dosing ([1](#), [2](#)). This approach may decrease nocturnal hypoglycemia ([3](#)).

Data to support the use of glucagon-like peptide-1 (GLP-1) receptor agonist therapy in the inpatient setting are limited ([1](#)).

Dipeptidyl peptidase-4 inhibitors (eg, sitagliptin, saxagliptin, linagliptin, alogliptin) are relatively safe, even in patients with kidney disease ([4](#)), and they may also be used for postprandial glucose lowering ([1](#)).

Hyperglycemia in critical illness

Inpatient hyperglycemia is associated with increased infection rate and mortality ([1](#), [5](#)). Critical illness causes insulin resistance and hyperglycemia even in patients without known diabetes mellitus. Such stress-induced hyperglycemia is associated with poor outcomes, including increased mortality. Insulin infusion to maintain plasma glucose between 140 and 180 mg/dL (7.8 and 10.0 mmol/L):

- Prevents adverse outcomes such as organ failure
- Possibly enhances recovery from stroke
- Leads to improved survival in patients requiring prolonged (> 5 days) critical care

It appears that maintaining less stringent insulin targets may be sufficient to prevent adverse outcomes (1). Severely ill patients, especially those treated with glucocorticoids or pressors and those receiving parenteral nutrition, may need very high doses of insulin (> 5 to 10 units/hour) because of insulin resistance. In critically ill patients, insulin infusion protocols and/or computerized algorithms can be used to titrate insulin drips to maintain euglycemia.

Patients undergoing surgery

A large percentage of patients undergoing surgery have diabetes or prediabetes (sometimes undiagnosed). The physiologic stress of surgery can increase plasma glucose in patients with diabetes and induce [diabetic ketoacidosis](#) in those with type 1 diabetes.

For shorter procedures, subcutaneous insulin can be used. In patients with type 1 diabetes, one-half of the usual morning dose of intermediate-acting insulin, or 75 to 80% of the dose of long-acting insulin (glargine or detemir) can be given the night or morning before surgery (at the usual time of long-acting insulin administration) (1).

During and after surgery, plasma glucose (and ketones if hyperglycemia suggests the need) should be measured at least every 2 to 4 hours (1). Dextrose infusion, or subcutaneous regular or short-acting insulin (every 4 to 6 hours as needed) can be given to maintain the plasma glucose level between 100 and 180 mg/dL (5.5 and 10.0 mmol/L) until the patient can be switched to oral feedings and resume the usual insulin regimen. Additional doses of intermediate- or long-acting insulin should be given if there is a substantial delay (> 24 hours) in resuming the usual regimen. This approach may also be used for insulin-treated patients with type 2 diabetes, but frequent measurement of ketones may be omitted.

Some physicians prefer to withhold subcutaneous or inhaled insulin on the day of surgery and to give insulin by IV infusion. For patients undergoing a long procedure or major surgery, a continuous insulin infusion is preferable, especially since insulin requirements can increase because of the stress of surgery. IV insulin infusion can be given at the same time as intravenous dextrose solution to maintain blood glucose. A common approach is to infuse insulin and dextrose separately. Insulin can be infused at a rate of 1 to 2 U/hour with 5% dextrose infusing at 75 to 150 mL/hour. The insulin rate may need to be decreased for patients with more insulin-sensitive type 1 diabetes and increased for patients with more insulin-resistant type 2 diabetes. An infusion of 10% dextrose may also be used. It is important, especially in patients with type 1 diabetes, to continue insulin infusion, even at a low rate, to avoid development of diabetic ketoacidosis.

Insulin adsorption onto IV tubing can lead to inconsistent effects, which can be minimized by preflushing the IV tubing with insulin solution. Insulin infusion is continued through recovery, with insulin dose adjusted based on the plasma glucose levels obtained in the recovery room and at 1- to 2-hour intervals thereafter. In postsurgical patients who are in an intensive care unit, insulin infusion protocols can be used to maintain euglycemia.

Insulin delivery through insulin pumps and automated insulin delivery systems can be continued during surgeries that last less than 2 hours (6). Continuous glucose monitors (CGMs) can also remain operational during surgery, but the accuracy of the measurements may be affected by fluoroscopy or electrocautery in the vicinity. Removal of pump insertion sets or CGMs may be required if they are within the surgical field. If a patient is off their insulin pump during surgery, intravenous insulin, or a dose of subcutaneous insulin can be administered to cover the period while the patient is off the pump. Intravenous or subcutaneous insulin administration is particularly important for patients with type 1 diabetes who should not have any periods without insulin coverage.

Non-insulin diabetes medications should be withheld before surgery. Metformin increases the risk of lactic acidosis and must be withheld the day of surgery. SGLT-2 inhibitors should be stopped 3 to 4 days before surgery to avoid diabetic ketoacidosis. GLP-1 receptor agonists can delay gastric emptying and increase the risk of aspiration in patients undergoing general anesthesia. Professional societies suggest that patients discontinue daily GLP-1 receptor agonists on the day of the procedure or surgery (1). Patients taking a once-weekly dose of a GLP-1 receptor agonist and those taking a dual glucose-dependent insulintropic polypeptide (GIP)/GLP-1 receptor agonist should not take these medications for at least 7 days prior to procedures or surgery (7). The HbA1C goal for elective surgeries should be < 8% (< 64.0 mmol/L).

Special populations references

1. [American Diabetes Association Professional Practice Committee](#). 16. Diabetes Care in the Hospital: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(Supplement_1):S321-S334. doi:10.2337/dc25-S016
2. [McCall AL, Lieb DC, Gianchandani R, et al](#). Management of Individuals With Diabetes at High Risk for Hypoglycemia: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2023;108(3):529-562. doi:10.1210/clinem/dgac596
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